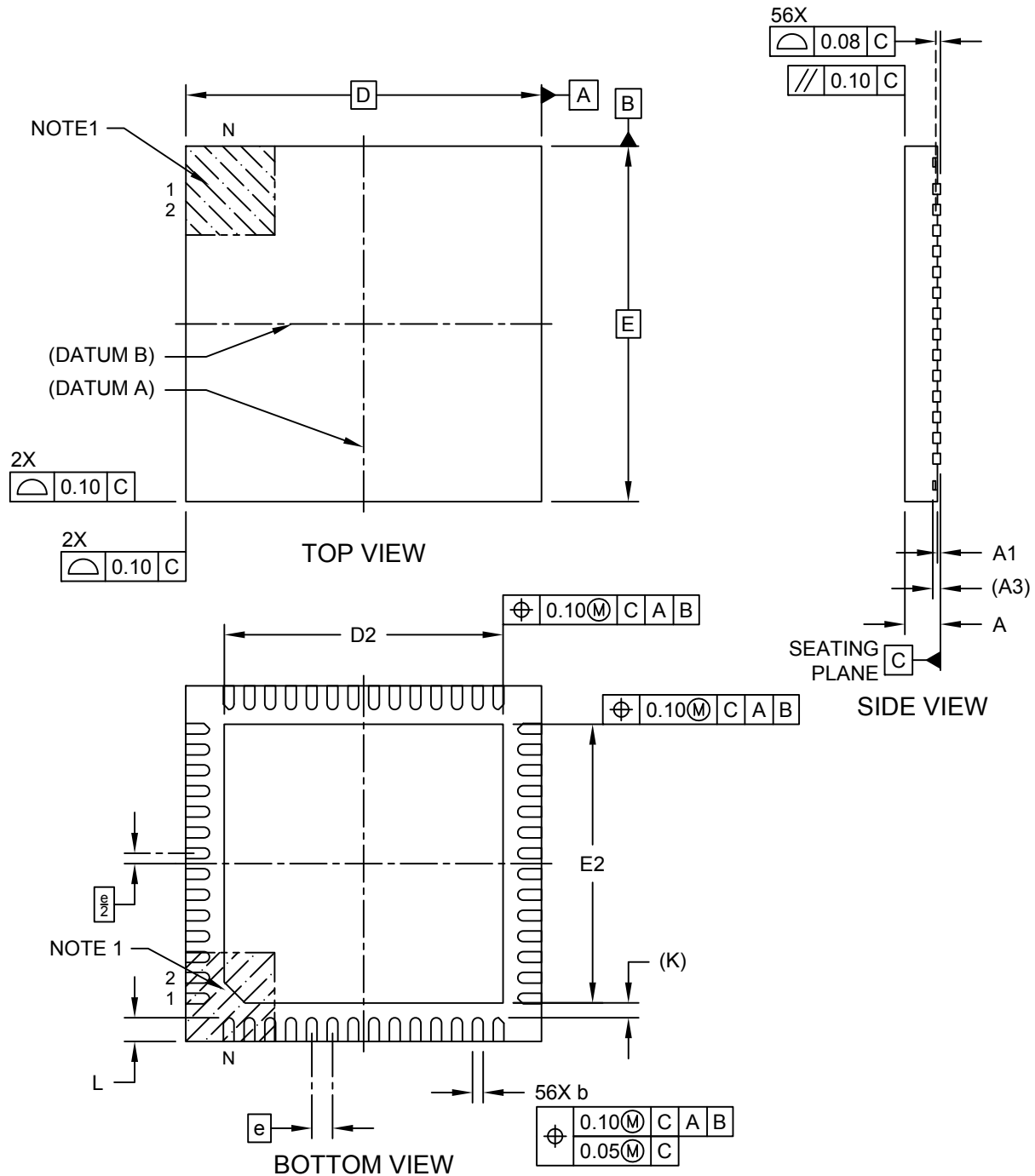


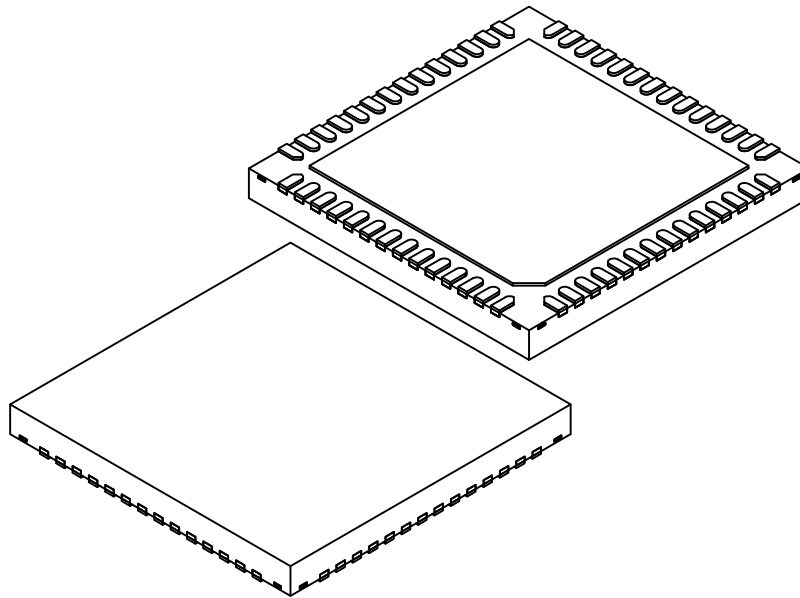
**56-Lead Ultra Thin Plastic Quad Flat, No Lead Package (R8B) - 6x6x0.6 mm Body [UQFN]  
With 4.7 mm Exposed Pad; Atmel Legacy Global Package Code ZTG**

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



**56-Lead Ultra Thin Plastic Quad Flat, No Lead Package (R8B) - 6x6x0.6 mm Body [UQFN]  
With 4.7 mm Exposed Pad; Atmel Legacy Global Package Code ZTG**

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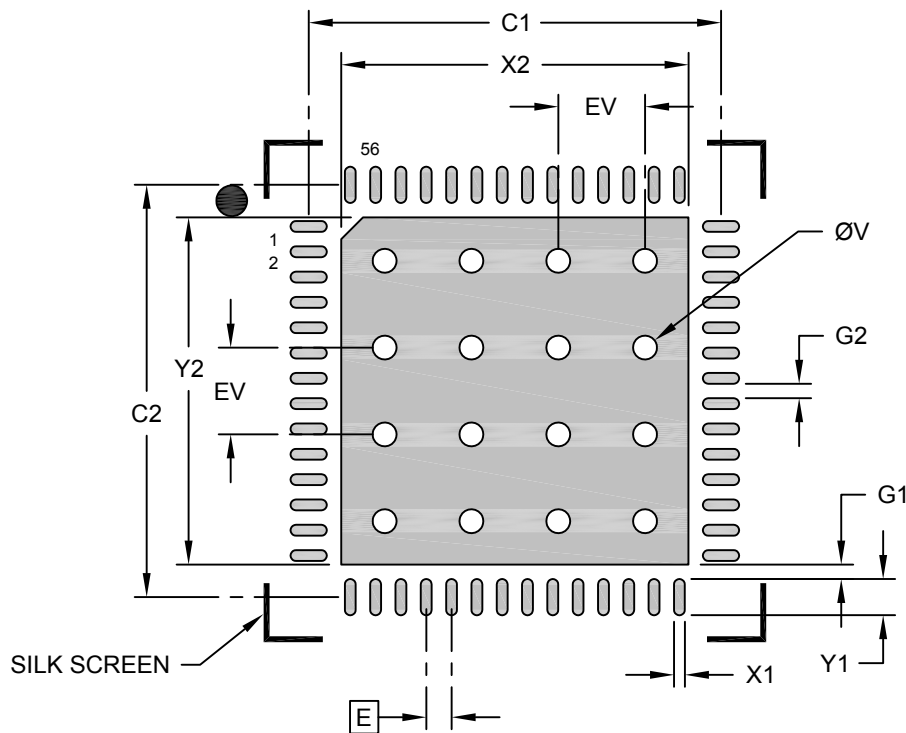
| Units                   |    | MILLIMETERS |      |      |
|-------------------------|----|-------------|------|------|
| Dimension Limits        |    | MIN         | NOM  | MAX  |
| Number of Terminals     | N  | 56          |      |      |
| Pitch                   | e  | 0.35 BSC    |      |      |
| Overall Height          | A  | -           | 0.55 | 0.60 |
| Standoff                | A1 | 0.00        | 0.02 | 0.05 |
| Terminal Thickness      | A3 | 0.127 REF   |      |      |
| Overall Length          | D  | 6.00 BSC    |      |      |
| Exposed Pad Length      | D2 | 4.60        | 4.70 | 4.80 |
| Overall Width           | E  | 6.00 BSC    |      |      |
| Exposed Pad Width       | E2 | 4.60        | 4.70 | 4.80 |
| Terminal Width          | b  | 0.13        | 0.18 | 0.23 |
| Terminal Length         | L  | 0.35        | 0.40 | 0.45 |
| Terminal-to-Exposed-Pad | K  | 0.25 REF    |      |      |

**Notes:**

1. Pin 1 visual index feature may vary, but must be located within the hatched area.
2. Package is saw singulated
3. Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.  
REF: Reference Dimension, usually without tolerance, for information purposes only.

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**RECOMMENDED LAND PATTERN**

| Units                            |    | MILLIMETERS |      |      |
|----------------------------------|----|-------------|------|------|
| Dimension Limits                 |    | MIN         | NOM  | MAX  |
| Contact Pitch                    | E  | 0.35 BSC    |      |      |
| Optional Center Pad Width        | X2 |             |      | 4.80 |
| Optional Center Pad Length       | Y2 |             |      | 4.80 |
| Contact Pad Spacing              | C1 |             | 5.70 |      |
| Contact Pad Spacing              | C2 |             | 5.70 |      |
| Contact Pad Width (X56)          | X1 |             |      | 0.15 |
| Contact Pad Length (X56)         | Y1 |             |      | 0.50 |
| Contact Pad to Center Pad (X56)  | G1 | 0.23        |      |      |
| Contact Pad to Contact Pad (X52) | G2 | 0.20        |      |      |
| Thermal Via Diameter             | V  |             | 0.33 |      |
| Thermal Via Pitch                | EV |             | 1.20 |      |

**Notes:**

- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-23374 Rev A